BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) — (Group-I) — HSSC-(P-I)-A/2024 (For All Sessions)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6197

Note: Note: Write Answers to the Questions on the objective answer sheet provided. Four possible answers A, B, C and D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with Marker or Pen ink on the answer sheet provided.

Q1. Four 4th roots of 625 are:

(A) ±4, ±4i

(B) ±5, ±5i

(C) ±16, ±16i

(D) ±25, ±25i

Q2. Partial fractions of $(x^2+1)/((x+1)(x-1))$ are of the form:

(A) $A/(x+1) + B/(x-1)$

(B) $Ax/(x+1) + B/(x-1)$

(C) $1 + A/(x+1) + B/(x-1)$

(D) $(Ax+B)/(x+1) + (Cx+D)/(x-1)$

Q3. A.M between $x-3$ and $x+5$ is:

(A) $x+1$

(B) $x-1$

(C) $x-3$

(D) $x+5$

Q4. No term of a G.P can be:

(A) 0

(B) 1

(C) -1

(D) i

Q5. 8.7.6 =

(A) 8!/8
(B) 8!/7!
(C) 8!/6!
(D) 8!/5!

Q6. $4^n > 3^n + 4$ is true for integers:

(A) $n \geq 2$
(B) $n \geq 3$
(C) $n \geq 4$
(D) $n \geq 5$

Q7. If $\sin\theta < 0$ and $\cos\theta > 0$, then terminal arm of θ lies in quadrant:

(A) I
(B) II
(C) III
(D) IV

Q8. $(1-\cos\theta)/2$ =

(A) $\sin\theta$
(B) $\sin^2(\theta/2)$
(C) $\cos\theta$
(D) $\cos^2(\theta/2)$

Q9. Range of y = tanx is:

(A) $-\pi/2 \leq y \leq \pi/2$
(B) $-\infty < y < \infty$
(C) [partially obscured by watermark on source scan]
(D) $-\infty < x < \infty$

Q10. 2R sin α = [(A)–(D) heavily watermarked on source scan; read as shown]

(A) r
(B) S
(C) Δ
(D) a

Q11. $\text{Sin}(\cos^{-1}(\sqrt{3}/2))$ =

(A) 1/2
(B) $\sqrt{3}/2$
(C) $1/\sqrt{3}$
(D) 1

Q12. Reference Angle for $1 - 2\sin x = 0$ is:

(A) π/6
(B) π/4
(C) π/3
(D) π/2

Q13. $\forall z \in C$, which one is true:

(A) $z = -z$
(B) $\bar{z} = -z$
(C) $\bar{\bar{z}} = z$
(D) $\bar{\bar{z}} = -z$

Q14. A prime number can be factor of a square only if it occurs in it at least:

(A) Once
(B) Twice
(C) Thrice
(D) Four times

Q15. If A and B are disjoint sets, then A − B =

(A) B (B) A

(C) B − A (D) φ

Q16. The converse of ~ p → q is:

(A) q → ~ p (B) p → q

(C) q → p (D) p → ~ q

Q17. p ∧ q is called:

(A) Conjunction (B) Disjunction

(C) Conditional (D) Equivalence

Q18. (AB)^t =

(A) A^t B^t (B) A^t B

(C) AB (D) B^t A^t

Q19. A square matrix A is anti-symmetric if:

(A) A^t = −A (B) A^t = A

(C) $\bar{A} = A$ (D) $\bar{A} = -A$

Q20. $1 + \omega + \omega^2 =$

(A) 1 (B) ω

(C) ω^2 (D) 0

Mathematics (Subjective) — (Group-I) — HSSC-(P-I)-A-2024 (For All Sessions)

Time Allowed: 2:30 hours | **Maximum Marks:** [illegible in source scan]

QSECTION-I.

Q2. (Marks (8x2=16)) Write short answers of any eight parts from the following:

i. Define a complex number. Is 0 a complex number?

ii. Whether the set {0, −1} is closed or not w.r.t. addition and multiplication.

iii. Factorize: $3x^2 + 3y^2$

iv. Find multiplicative inverse of −3 − 5i

v. Construct truth table of ~(p → q) → p

vi. Define monoid.

vii. Find the matrix X if: X [[5, 2], [−2, 1]] = [[−1, 5], [12, 3]]

viii. If A and B are square matrices of the same order, then explain why in general $(A + B)^2 \neq A^2 + 2AB + B^2$

ix. If A = [1/(1+i), i]^t, find A(Ā)^t

x. Find four fourth roots of 81

xi. Use the remainder theorem to find the remainder when $x^3 - 2x^2 + 3x + 3$ is divided by $x - 3$

xii. If α, β are the roots of $3x^2 - 2x + 4 = 0$, find the value of $\alpha/\beta + \beta/\alpha$

Q3. (Marks (8x2=16)) Write short answers of any eight parts from the following:

i. Define conditional equation.

ii. Resolve $(x^2+15)/((x^4+2x+5)(x-1))$ into partial fraction without finding constants.

iii. Find the first four terms of the sequence $a_n = n/(2n+1)$

iv. Determine whether −19 is a term of 17, 13, 9, ...

v. Find the 5th term of the G.P 3, 6, 12, ...

vi. Sum the series $3/\sqrt{2} + 2\sqrt{2} + 5/\sqrt{2} + ... + a_{13}$

vii. Prove from the first principle that ${}^nP_r = n \cdot {}^{n-1}P_{r-1}$

viii. Find the value of n when ${}^nC_{12} = {}^nC_6$

ix. Determine the probability of getting dots less than 5 when a die is rolled.

x. Prove that $n! > 2^n - 1$ for n = 4, 5

xi. Calculate $(2.02)^4$ by means of binomial theorem.

xii. Expand $(1 + 2x)^{-1}$ up to 4 terms.

Q4. (Marks (9x2=18)) Write short answers of any nine parts from the following:

i. Write values of trigonometric functions for $\theta = (-9/2)\pi$.

ii. Prove that $t^2\theta - \cos^2\theta = \cot^2\theta \cos^2\theta$.

iii. Prove that $\sin(\theta + 270) = -\cos\theta$.

iv. Prove that $\sin2\theta = 2\sin\theta \cos\theta$.

v. Express sin12° sin46° as sum or difference.

vi. Write domain and range of cos x.

vii. Find period of sin(x/3).

viii. Draw the graph of tanx for $x \in (0, \pi)$

ix. Prove that $r = (s - b)\tan(\beta/2)$.

x. Write any two half angle formulae.

xi. When angle between ground and sun is 30°, flag pole casts a shadow of 40m long. Find height of top of flag.

xii. Show that $\cos(\sin^{-1}x) = \sqrt{(1 - x^2)}$.

xiii. Solve the equation $4\cos^2x - 3 = 0$

QSECTION-II.

Note: Attempt any three questions. Each question carries equal marks: (10x3=30)

Q5. (a) If α and β are the roots of $x^2 - 3x + 5 = 0$, form the equation whose roots are $(1-\alpha)/(1+\alpha)$ and $(1-\beta)/(1+\beta)$ (b) Find the rank of matrix [[1, −1, 2, 1], [2, −6, 5, 1], [3, 5, 4, −3]]

Q6. (a) Resolve $1/((x-1)^2(x^2+2))$ into partial fractions. (b) Find six arithmetic means between 2 and 5.

Q7. (a) A die is thrown. Find the probability that the no. of dots on the top are prime numbers or odd numbers. (b) If x is so small that its cube or higher powers can be neglected, show that $\sqrt{((1-x)/(1+x))} \approx 1 - x + (1/2)x^2$

Q8. (a) Solve the triangle ABC, given that α = 35° 17' β = 45° 13', b = 421. (b) Reduce $\cos^4\theta$ to an expression involving only function of multiples of θ, raised to the first power.

Q9. (a) A circular wire of radius 6 cm is cut straightened and then bent so as to lie along the circumference of a hoop of radius 24 cm. Find the measure of the angle which it subtends at the center of the hoop. (b) Prove that: $\tan^{-1}(1/4) + \tan^{-1}(1/5) = \tan^{-1}(9/19)$

GROUP – II

Mathematics (Objective) — (Group-II) — HSSC-(P-I)-A/2024 (For All Sessions)

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 6192

Note: Note: Write Answers to the Questions on the objective answer sheet provided. Four possible answers A, B, C and D to each question are given. Which answer you consider correct, fill the corresponding circle A, B, C or D given in front of each question with Marker or Pen ink on the answer sheet provided.

Q1. A complex number 1 + i can also be expressed as:

(A) $2(\cos 45° + i \sin 45°)$ (B) $\sqrt{2}(\cos 45° - i \sin 45°)$

(C) $\sqrt{2}(\cos 45° + i \sin 45°)$ (D) $2(\cos 45° - i \sin 45°)$

Q2. If Z is a complex number and $Z = \bar{Z}$ then Z must be:

(A) Real (B) Imaginary

(C) Rational (D) Irrational

Q3. The set {(a, b)} is called:

(A) Infinite set (B) Singleton set

(C) Empty set (D) Set with two elements

Q4. Drawing conclusion from premises believed to be true is called:

(A) Proposition (B) Contradiction

(C) Induction (D) Deduction

Q5. If p is a logical statement $p \wedge \sim p$ is always:

(A) Absurdity (B) Contigency [sic, as printed]

(C) Tautology (D) Conditional

Q6. If A = [a b c], then order of A^t is:

(A) 1×3 (B) 3×1

(C) 3×3 (D) 1×1

Q7. If the matrix $[[\lambda, 1], [-2, 1]]$ is singular then $\lambda =$

(A) 2 (B) 1

(C) -1 (D) -2

Q8. IF 4^{3x} = 1/2 then x is equal to:

(A) $-1/6$ (B) -6

(C) $1/6$ (D) 6

Q9. If ω is cube root of unity, then $\omega + \omega^2 =$

(A) 0 (B) -1

(C) 1 (D) $1/\omega$

Q10. From the identity $5x + 4 = A(x-1) + B(x+2)$, value of B is:

(A) -3 (B) 3

(C) -2 (D) 2

Q11. Which of the term cannot be a term of G.P:

(A) -1 (B) 1

(C) 0 (D) 5

Q12. $\Sigma_{k=1}^{n} K$ is equal to:

(A) $(n+1)/2$ (B) $n(n+1)/2$

(C) $n(n+1)(2n+1)/6$ (D) $n(n-1)/2$

Q13. ${}^{n}P_r / r!$ is equal to:

(A) ${}^{n}C_r$ (B) ${}^{n}C_{r-1}$

(C) ${}^{n+1}C_r$ (D) ${}^{n-1}C_r$

Q14. In expansion of $(a + b)^{16}$ middle term will be:

(A) 11th (B) 12th

(C) 8th (D) 9th

Q15. Which of the following is NOT Quadrantal angle?

(A) $(9/2)\pi$ (B) 13π

(C) $(4/3)\pi$ (D) $\pi/2$

Q16. The angle $(3\pi/2) - \theta$ lies in quadrant:

(A) I (B) II

(C) III (D) IV

Q17. The range of sinx is:

(A) $[-1, 1]$ (B) $[-1, 0]$

(C) $[0, 2]$ (D) $[-2, 2]$

Q18. The radius of inscribed circle is:

(A) $abc/(4\Delta)$ (B) S/Δ

(C) $\Delta/(S-a)$ [partially obscured by watermark on source scan] (D) Δ/S

Q19. $\text{Cos}(\sin^{-1}(1/\sqrt{2}))$ is equal to:

(A) 1/2 (B) $\pi/4$

(C) $1/\sqrt{2}$ (D) $-\pi/4$

Q20. If $\sin x = 1/2$, then reference angle is:

(A) $\pi/3$ (B) $\pi/4$

(C) $-\pi/6$ (D) $\pi/6$

Mathematics (Subjective) — (Group-II) — HSSC-(P-I)-A-2024 (For All Sessions)

Time Allowed: 2:30 hours | **Maximum Marks:** 80

QSECTION-I.

Q2. (Marks (8x2=16)) Write short answers of any eight parts from the following:

i. Does the set {1, −1} possess closure property w.r.t. multiplication? Construct the multiplication table.

ii. If a/b = c/d, prove that ad = bc

iii. Factorize $a^2 + 4b^2$

iv. Simplify by expressing in the form a + bi: $(2 + \sqrt{(-3)})(3 + \sqrt{(-3)})$

v. If B = {1, 2, 3} then write down the power set of B

vi. Determine whether the statement $p \rightarrow (q \rightarrow p)$ is a tautology or not.

vii. Under what conditions, the determinant of a square matrix A is zero. Write any two conditions.

viii. If A = [[1, 2], [a, b]] and A^2 = [[0, 0], [0, 0]], find the values of a and b.

ix. Determine whether the matrix A = [[1, 1+i], [1−i, 2]] is hermitian matrix or skew-hermitian matrix.

x. Solve the equation: $x^{-2} - 10 = 3x^{-1}$

xi. Find four fourth roots of 16.

xii. Show that the roots of equation will be rational $px^2 - (p - q)x - q = 0$

Q3. (Marks (8x2=16)) Write short answers of any eight parts from the following:

i. Define an identity with example.

ii. Resolve into partial fraction $1/(x^2-1)$

iii. The 7th and 10th terms of an H.P are 1/3 and 5/21 respectively, find its 14th term.

iv. Find the sum of first 15 terms of geometric sequence 1, 1/3, 1/9, ...

v. Insert two G.M's between 2 and 16.

vi. How many terms of the series −7 + (−5) + (−3) + ... amount to 65

vii. A card in drawn [sic, as printed] from a deck of 52 playing cards. What is the probability that it is a diamond card or an ace?

viii. Find n, if ${}^nC_8 = {}^nC_{12}$

ix. How many different 4-digit numbers can be formed out of the digits 1, 2, 3, 4, 5, 6, when no digit is repeated?

x. Use mathematical induction to prove that $3 + 3.5 + 3.5^2 + ... + 3.5^n = 3(5^{n+1}-1)/4$ for n = 1, 2

xi. Calculate by means of binomial theorem $(2.02)^4$

xii. Expand upto 4 − terms (1 − x)^(1/2)

Q4. (Marks (9x2=18)) Write short answers of any nine parts from the following:

i. Find r, when l = 56cm, θ = 45°

ii. Verify that $\sin2\theta = 2\sin\theta\cos\theta$ for θ = 45°

iii. Write the fundamental law of trigonometry.

iv. Show that $\cos(\alpha+\beta)\cos(\alpha-\beta) = \cos^2\alpha - \sin^2\beta$.

v. Express sin5x + sin7x as a product.

vi. Define the period of trigonometric function.

vii. Write down the domain and range of tangent function.

viii. Find the period of sin(x/3)

ix. Solve the right triangle ABC, in which γ = 90°, a = 3.28, b = 5.74.

x. Define half angle formulas for tangent.

xi. Define Hero's formula.

xii. Find the value of $\sin(\tan^{-1}(-1))$

xiii. Solve the equation sin2x = cosx where x ∈ [0, 2π]

QSECTION-II.

Note: Attempt any three questions. Each question carries equal marks: (10x3=30)

Q5. (a) Show that |x 1 1 1; 1 x 1 1; 1 1 x 1; 1 1 1 x| = (x + 3)(x − 1)³ (b) Prove that $x^2/a^2 + (mx+c)^2/b^2 = 1$ will have equal roots if $c^2 = a^2m^2 + b^2$; a ≠ 0, b ≠ 0

Q6. (a) Resolve into partial fractions $(6x^3+5x^2-7)/(2x^2-x-1)$ (b) The A.M between the two numbers is 5 and their positive G.M. is 4 find the numbers.

Q7. (a) Prove that ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$ (b) Find the coefficient of x^5 in the expansion of $(x^2 - 3/(2x))^{10}$

Q8. (a) Reduce $\sin^4\theta$ to an expression involving only functions of multiples of θ raised to the first power. (b) With usual notations, prove that r = s.tan(α/2).tan(β/2).tan(γ/2)

Q9. (a) If $\cot\theta = 5/2$, and θ is in quadrant I, find the value of $(3\sin\theta+4\cos\theta)/(\cos\theta-\sin\theta)$ (b) Prove that $\cos^{-1}(63/65) + 2\tan^{-1}(1/5) = \sin^{-1}(3/5)$